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Bagging and Random Forest Ensemble Algorithms for Machine Learning

by **Jason Brownlee** on December 3, 2020 in **Machine Learning Algorithms** 133

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Random Forest is one of the most popular and most powerful machine learning algorithms. It is a type of ensemble machine learning algorithm called Bootstrap Aggregation or Bagging.

Ensemble Selection for Bagging for Machine Learning

In this post you will discover the Bagging ensemble algorithm and the Random Forest algorithm for predictive modeling. After reading this post you will know about:

Logistic Regression for Machine Learning

- The bootstrap method for estimating statistical quantities from samples.
- The Bootstrap Aggregation algorithm for creating multiple different models from a single training dataset.
- The Random Forest algorithm that makes a small tweak to Bagging and results in a very powerful classifier.

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Bagging and Random Forest Ensemble Algorithms for Machine Learning
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Bootstrap Method

Before we get to Bagging, let's take a quick look at an important foundation technique called the bootstrap.

The bootstrap is a powerful statistical method for estimating a quantity from a data sample. This is easiest to understand if the quantity is a descriptive statistic such as a mean or a standard deviation.

Let's assume we have a sample of 100 values (x) and we'd like to get an estimate of the mean of the sample.

We can calculate the mean directly from the sample as:

$$\text{mean}(x) = 1/100 * \text{sum}(x)$$

We know that our sample is small and that our mean has error in it. We can improve the estimate of our mean using the bootstrap procedure:

1. Create many (e.g. 1000) random sub-samples of our dataset with replacement (meaning we can select the same value multiple times).
2. Calculate the mean of each sub-sample.
3. Calculate the average of all of our collected means and use that as our estimated mean for the data.

For example, let's say we used 3 resamples and got the mean values 2.3, 4.5 and 3.3. Taking the average of these we could take the estimated mean of the data to be 3.367.

This process can be used to estimate other quantities like the standard deviation and even quantities used in machine learning algorithms, like learned coefficients.

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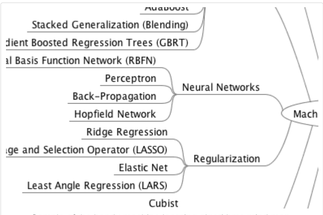

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Bootstrap Aggregation (Bagging)

Bootstrap Aggregation (or Bagging for short), is a simple and very powerful ensemble method.

An ensemble method is a technique that combines the predictions from multiple machine learning algorithms together to make more accurate predictions than any individual model.

Bootstrap Aggregation is a general procedure that can be used to reduce the variance for those algorithm that have high variance. An algorithm that has high variance are decision trees, like classification and regression trees (CART).

Decision trees are sensitive to the specific data on which they are trained. If the training data is changed (e.g. a tree is trained on a subset of the training data) the resulting decision tree can be quite different and in turn the predictions can be quite different.

Bagging is the application of the Bootstrap procedure to a high-variance machine learning algorithm, typically decision trees.

Let's assume we have a sample dataset of 1000 instances (x) and we are using the CART algorithm. Bagging of the CART algorithm would work as follows.

1. Create many (e.g. 100) random sub-samples of our dataset with replacement.
2. Train a CART model on each sample.
3. Given a new dataset, calculate the average prediction from each model.

For example, if we had 5 bagged decision trees that made the following class predictions for a in input sample: blue, blue, red, blue and red, we would take the most frequent class and predict blue.

When bagging with decision trees, we are less concerned about individual trees overfitting the training data. For this reason and for efficiency, the individual decision trees are grown deep (e.g. few training samples at each leaf-node of the tree) and the trees are not pruned. These trees will have both high variance and low bias. These are important characteristics of sub-models when combining predictions using bagging.

The only parameters when bagging decision trees is the number of samples and hence the number of trees to include. This can be chosen by increasing the number of trees on run after run until the accuracy begins to stop showing improvement (e.g. on a cross validation test harness). Very large numbers of models may take a long time to prepare, but will not overfit the training data.

Just like the decision trees themselves, Bagging can be used for classification and regression problems.

Random Forest

Random Forests are an improvement over bagged decision trees.

A problem with decision trees like CART is that they are greedy. They choose which variable to split on using a greedy algorithm that minimizes error. As such, even with Bagging, the decision trees can have a lot of structural similarities and in turn have high correlation in their predictions.

Combining predictions from multiple models in ensembles works better if the predictions from the sub-models are uncorrelated or at best weakly correlated.

Random forest changes the algorithm for the way that the sub-trees are learned so that the resulting predictions from all of the subtrees have less correlation.

It is a simple tweak. In CART, when selecting a split point, the learning algorithm is allowed to look through all variables and all variable values in order to select the most optimal split-point. The random forest algorithm changes this procedure so that the learning algorithm is limited to a random sample of features of which to search.

The number of features that can be searched at each split point (m) must be specified as a parameter to the algorithm. You can try different values and tune it using cross validation.

- For classification a good default is: m = sqrt(p)
- For regression a good default is: m = p/3

Where m is the number of randomly selected features that can be searched at a split point and p is the number of input variables. For example, if a dataset had 25 input variables for a classification problem, then:

- m = sqrt(25)
- m = 5

For each bootstrap sample taken from the training data, there will be samples left behind that were not included. These samples are called Out-Of-Bag samples or OOB.

The performance of each model on its left out samples when averaged can provide an estimated accuracy of the bagged models. This estimated performance is often called the OOB estimate of performance.

These performance measures are reliable test error estimate and correlate well with cross validation estimates.

Variable Importance

As the Bagged decision trees are constructed, we can calculate how much the error function drops for a variable at each split point.

In regression problems this may be the drop in sum squared error and in classification this might be the Gini score.

These drops in error can be averaged across all decision trees and output to provide an estimate of the importance of each input variable. The greater the drop when the variable was chosen, the greater the importance.

These outputs can help identify subsets of input variables that may be most or least relevant to the problem and suggest at possible feature selection experiments you could perform where some features are removed from the dataset.

Further Reading

Bagging is a simple technique that is covered in most introductory machine learning texts. Some examples are listed below.

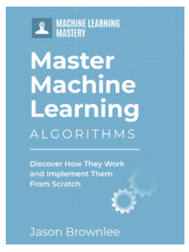
- An Introduction to Statistical Learning: with Applications in R, Chapter 8.
- Applied Predictive Modeling, Chapter 8 and Chapter 14.
- The Elements of Statistical Learning: Data Mining, Inference, and Prediction, Chapter 15

Summary

In this post you discovered the Bagging ensemble machine learning algorithm and the popular variation called Random Forest. You learned:

- How to estimate statistical quantities from a data sample.
- How to combine the predictions from multiple high-variance models using bagging.
- How to tweak the construction of decision trees when bagging to de-correlate their predictions, a technique called Random Forests.

Do you have any questions about this post or the Bagging or Random Forest Ensemble algorithms?
Leave a comment and ask your question and I will do my best to answer it.



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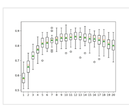
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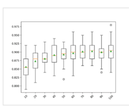
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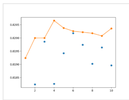
How to Create a Bagging Ensemble of Deep Learning…

About Jason Brownlee

Jason Brownlee, PhD is a machine learning specialist who teaches developers how to get results with modern machine learning methods via hands-on tutorials.
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133 Responses to *Bagging and Random Forest Ensemble Algorithms for Machine Learning*

Maria July 9, 2016 at 12:21 am #

Thanks for your clear and helpful explanation of bagging and random forest. I was just wondering if there is any formula or good default values for the number of models (e.g., decision trees) and the number of samples to start with, in bagging method? Is there any relation between the size of training dataset (n), number of models (m), and number of sub-samples (n') which I should obey?

Jason Brownlee July 9, 2016 at 7:41 am #

Great questions Maria, I'm not aware of any systematic studies off the top of my head.

A good heuristic is to keep increasing the number of models until performance levels off.

Also, it is generally a good idea to have sample sizes equal to the training data size.

Alvie May 23, 2018 at 1:05 pm #

Hi Jason, if the sample size equal to the training data size, how there are out of bag samples? I am so confused about this.

Jason Brownlee May 23, 2018 at 2:40 pm #

Because we are selecting examples with replacement, meaning we are including some examples many times and the sample will likely leave many examples that were not included.

Does that help?

Alvie May 23, 2018 at 4:10 pm #

Hi Jason. Thank you so much!

Hamza November 13, 2018 at 4:58 pm #

You need to pick data with replacement. This mean if sample data is same training data this mean the training data will increase for next smoking because data picked twice and triple and more. The best thing is pick 60% for training data from sample data to make sure variety of output will occurred with different results. I repeat. Training data must be less than sample data to create different tree construction based on variety data with replacement.

Anh Hoang Duc September 12, 2016 at 2:32 pm #

Hi @Maria,
In R, you can use function tuneRF in randomForest package to find optimal parameters for randomForest

Luis O October 15, 2016 at 2:31 am #

I always read your posts @Jason Brownlee. However I thinkt that in this case, you would need some figures to explain better. Also, try to use different font style when you are refering to formulas.

Jason Brownlee October 15, 2016 at 10:25 am #

Thanks for the feedback Luis, much appreciated.

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@Jason – Can I know in case of baggaing and boosting, we use multiple algorithms (e.g. decison tree, Logistic regression, SVM etc) or just any single algorithm to produce multiple models?

REPLY

Jason Brownlee January 6, 2020 at 7:10 am #

In bagging and boosting we typically use one algorithm type and traditionally this is a decision tree.

REPLY

Santhosh January 6, 2020 at 9:00 pm #

@Jason Brownlee can u Elaborate all concepts in machine learning with real time examples?

REPLY

Jason Brownlee January 6, 2020 at 7:11 am #

That is what this blog covers.

REPLY

OpenFlow December 11, 2016 at 8:39 pm #

Hey Jason,

You're doing a great job here. I didn't know anything about machine learning until I found your site. Believe it or not, I follow it pretty well. #LoveMath.

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REPLY

Jason Brownlee December 2, 2016 at 8:11 am #

REPLY

Shilpa Anil January 4, 2017 at 4:52 am #

Sir, I have to predict daily air temperature values using random forest regression and i have 5 input varibales. Could you please explain how splitting is performed in regression? exactly what is done at each split point?

REPLY

Jason Brownlee January 4, 2017 at 8:59 am #

Here is some advice on splitting time series data for machine learning:
https://machinelearningmastery.com/backtest-machine-learning-models-time-series-forecasting/

REPLY

Ola January 13, 2017 at 3:14 am #

Thank You for that post! It helps me to clarify decision about using Random Forest in my Master's Thesis analysis.

REPLY

Jason Brownlee January 13, 2017 at 9:14 am #

I'm glad to hear it.

REPLY

Joe Ellsworth January 24, 2017 at 1:06 pm #

Hi Jason, I liked your article. I am working on a Quantized classifier and would love to collaborate on an article. https://bitbucket.org/joexdobs/ml-classifier-gesture-recognition

REPLY

Jason Brownlee January 25, 2017 at 9:57 am #

Good luck with your project Joe!

REPLY

Chandrakant Pattekar May 28, 2017 at 6:49 pm #

Very well explained

REPLY

Jason Brownlee June 2, 2017 at 12:10 pm #

Thanks, I'm glad it helped.

REPLY

Mercy Peter June 23, 2017 at 3:26 pm #

Very crisp and clear explanations, nailed to the point.

REPLY

Jason Brownlee June 24, 2017 at 7:58 am #

Thanks Mercy.

REPLY

Paul C June 28, 2017 at 1:35 am #

Hi Jason, it's not true that bootstrapping a sample and computing the mean of the bootstrap sample means "improves the estimate of the mean." The standard MLE (I.e just the sample mean) is the best estimate of the population mean. Bootstrapping is great for many things but not for giving a better estimate of a mean.

REPLY

Jason Brownlee June 28, 2017 at 6:29 am #

Perhaps. Also, check this:
https://en.wikipedia.org/wiki/Bootstrapping_(statistics)#Estimating_the_distribution_of_sample_mean

REPLY

Felipe July 30, 2017 at 1:49 am #

Hi, Jason! I'm reading your article and helped me understand the context about bagging. I need to implement a Bagging for Caltech 101 dataset and I do not know how can I start. I am programing somenthing in Matlab but I dont know how can I create a file from Caltech101 to Matlab and studying the data to create Ensemble. I am little confusing! Do you have any consideration to help me?

REPLY

Jason Brownlee July 30, 2017 at 7:49 am #

Sorry, I do not have matlab examples. I may not the best person to give you advice.

REPLY

Sonal August 8, 2017 at 12:43 am #

Hi Jason, Can you recommend any C++ libraries (open source or commercially licensed) with an accurate implementation of decision trees and its variants(bagged, random forests)?

REPLY

Jason Brownlee August 8, 2017 at 7:50 am #

Not really. Perhaps xgboost – I think it is written in cpp.

REPLY

Himanshu November 17, 2017 at 2:24 am #

When label data is very less in my training how can I use bagging to validate performance on the full distribution of training?

REPLY

Jason Brownlee November 17, 2017 at 9:27 am #

Not sure I follow, can you elaborate?

REPLY

Himanshu November 17, 2017 at 9:23 pm #

I mean out of 100k training data I have 2k labeled, so can I use bagging to label rest of my unlabeled data in training data set, I will do cross validation before bagging within 2k labelled

REPLY

Jason Brownlee November 18, 2017 at 10:16 am #

You could build a model on the 2K and predict labels for the remaining 100k, and you will need to test a suite of methods to see what works best using cross validation.

REPLY

Rajit Nair December 4, 2017 at 5:08 pm #

Hi Jason, Your blogs are always very useful to me, but it will be more useful when you take an example and explain the whole process. But anyways you blogs are very new and interesting

REPLY

Jason Brownlee December 5, 2017 at 5:41 am #

Thanks Rajit.

REPLY

Simon Ting December 26, 2017 at 9:45 pm #

Nice tutorial, Jason! I'm a bit confuse about the "Variable Importance" part, which step in bagging algorithm do you need to calculate the importance of each variables by estimate the error function drops?

REPLY

Jason Brownlee December 27, 2017 at 5:19 am #

Thanks.

I'm not sure I follow, perhaps you can restate the question?

REPLY

sachindeo chavan January 14, 2018 at 10:33 am #

Dear Jason, I'm new to regression am a student of MSc Big Data Analytics Uinversity of Liverpool UK. My query is on Random Forest, is Random Forest non-parametric regression model? regards sachin

REPLY

Jason Brownlee January 15, 2018 at 6:55 am #

Yes.

REPLY

Alejandra January 23, 2018 at 6:56 pm #

Thanks for making it clear. I have not enough background (I am a journalist) and was easy to understand. Thanks for sharing your knowledge!

REPLY

Jason Brownlee January 24, 2018 at 9:52 am #

REPLY

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Josie January 28, 2018 at 4:35 am #

Great post! Very clearly explained bagging and Random Forest. Very helpful. Thank you so much!

REPLY

Picked for you:

Jason Brownlee January 28, 2018 at 8:26 am #

You're welcome, glad it helped.

REPLY

hemant yadav March 6, 2018 at 10:09 pm #

I cannot say how helpful this post is to me. thanks for posting this.

REPLY

Jason Brownlee March 7, 2018 at 6:13 am #

I'm glad to hear that.

REPLY

Hamed S March 27, 2018 at 2:59 am #

THANKS for the helpful article, Jason!

I think in the following phrase 'sample' should be replaced with 'population':

Let's assume we have a sample of 100 values (x) and we'd like to get an estimate of the mean of the 'sample'.

It's been also stated in Wikipedia:

"The basic idea of bootstrapping is that inference about a population from sample data . . . "

REPLY

Jason Brownlee March 27, 2018 at 6:40 am #

Correct, we estimate population parametres using data samples.

REPLY

VISHESH SHARMA May 17, 2018 at 12:16 am #

Hey Jason, thanks for the post.

Wanted a clarification:

1. Bagging is row subsampling not feature/column subsampling?

2. In Random Forest, feature subsampling is done at every split or for every tree?

3. Random Forest uses both bagging (row sub sampling) and feature subsampling?

Regards,

REPLY

Jason Brownlee May 17, 2018 at 6:34 am #

Yes, row sampling.

Yes, feature sampling is performed at each split point.

Yes, it does both.

REPLY

mohammed musthafa June 13, 2018 at 5:48 pm #

can we use this method for predicting some numerical value or is it only for classification

REPLY

Jason Brownlee June 14, 2018 at 5:59 am #

Yes, this model could be used for regression.

REPLY

joe June 15, 2018 at 3:00 pm #

what is the difference between bagging and random forest?

REPLY

Jason Brownlee June 16, 2018 at 7:24 am #

Good question.

Bagging will use the best split point to build trees from a random subsample of the dataset.

RF will use the whole dataset but will choose the best split points in trees using a random subset of features in the dataset

REPLY

Pham Dat August 1, 2018 at 1:14 pm #

Hi Jason;

Could you please explain for me what is the difference between random forest, rotation forest and deep forest?
Do you implement rotation forest and deep forest in Python or Weka Environment? If so, please send the link. Many thanks

REPLY

Jason Brownlee August 1, 2018 at 2:22 pm #

Good question, I'm not sure off the cuff.

REPLY

Tanweer August 9, 2018 at 12:56 pm #

Very well explained in layman term. Many thanks

REPLY

Jason Brownlee August 9, 2018 at 2:05 pm #

Thanks. I'm happy it helped.

REPLY

Asm August 10, 2018 at 1:45 am #

Hi Jason,

Is it safe to say that Bagging performs better for binary classification than for multiple classification?
if that is so, why?

REPLY

Jason Brownlee August 10, 2018 at 6:20 am #

It is problem specific.

REPLY

Leon C August 15, 2018 at 12:52 am #

Hi Jason,
You mentioned "As such, even with Bagging, the decision trees can have a lot of structural similarities and in turn have high correlation in their predictions."

Why is high correlation bad in this case? Can you please give me an example?

REPLY

Jason Brownlee August 15, 2018 at 6:05 am #

Ensembles are more effective when their predictions (errors) are uncorrelated/weakly correlated.

REPLY

Vivek October 2, 2018 at 7:59 pm #

Hi Jason, great article.I have a confusion though.

As you mentioned in the post, a submodel like CART will have low bias and high variance. The meta bagging model(like random forest) will reduce the reduce the variance. Since, the submodels already have low bias, I am assuming the meta model will also have low bias. However, I have seen that it generally gets stated that bagging reduces variance, but not much is mentioned about it giving a low bias model as well. Could you please explain that?

Also, if bagging gives models with low bias and reduces variance(low variance) , than why do we need boosting algorithms?

REPLY

Jason Brownlee October 3, 2018 at 6:17 am #

No the sub models have low bias and higher variance, the bagged model has higher bias and lower variance.

This is the beauty of the approach, we can get a _usefully_ higher bias by combining many low bias models.

Boosting achieves a similar result a completely different way. We need many approaches as no single approach works well on all problems.

REPLY

Anitya October 9, 2018 at 1:04 pm #

for each sample find the ensemble estimate by finding the most common prediction (the mode)?

Compute the accuracy of the method by comparing the ensemble estimates to the truth?

REPLY

Jason Brownlee October 9, 2018 at 3:05 pm #

Sorry, I don't follow, can you elaborate your question?

REPLY

Shenghuan December 6, 2018 at 9:36 am #

Hi Jason,

Thanks for your good article.
I have a question that for each node of one tree, do they search in the same sub-set features? Or for each node, the program searches a new sub-set features?

Thanks again.

REPLY

Jason Brownlee December 6, 2018 at 1:42 pm #

A new subset is created and searched at each spit point.

REPLY

JinG January 22, 2019 at 1:05 am #

Hi Jason, by "subsamples with replacement', do you mean a single row can apear multiple times in one of the subsample? Or it can not but it can apear in multiple subsamples? Thanks.

REPLY

Jason Brownlee January 22, 2019 at 6:24 am #

It can appear multiple times in one sample.

REPLY

vaid February 1, 2019 at 5:01 pm #

so does it mean one row can appear multiple time in single tree..i.e. if i have rows x1,x2..xn..lets say x1 appear 2 times in first tree and x1,x2 appear 4 times in second tree for random forest

REPLY

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A split point uses one value for one feature. Different values for the same or different features can be reused, even the same value for the same feature – although I doubt it.

REPLY

Ken Popkin February 10, 2019 at 2:36 am #

Hi Jason,

ik I understand this post, but I'm getting confused as I read up on ensembles. Specifically, is applying them…

option 1: as simple as just choosing to use an ensemble algorithm (I'm using Random Forest and AdaBoost)

n 2: is it more complex, i.e. am I supposed to somehow take the results of my other algorithms (I'm using Logistic Regression, KNN, and Naïve-Bayes) and ehow use their output as input to the ensemble algorithms.

I think it's option 1, but as mentioned above some of the reading I've been doing is confusing me.

nks,

REPLY

Jason Brownlee February 10, 2019 at 9:45 am #

What is confusing you exactly?

REPLY

Venu February 27, 2019 at 1:02 am #

Hi

So it means each tree in the random forest will have low bias and high variance?

Are high variance and low bias?

Thanks,

Venu

REPLY

Jason Brownlee February 27, 2019 at 7:33 am #

Correct!

REPLY

Aisha May 11, 2019 at 12:56 am #

Hello sir,

Thanks for your article.
I have a question about time series forecasting with bagging.

In fact my base is composed of 500 days, each day is a time series (database: 24 lines (hours), 500 columns (days))
I want to apply a bagging to predict the 501 day.
How can i apply this technique given it resamples the base into subsets randomly and each subset makes one-day forecasting at random.
Is the result of the aggregation surely the 501 day?

thank you

REPLY

Jason Brownlee May 11, 2019 at 6:18 am #

This post will help to frame your data:
https://machinelearningmastery.com/convert-time-series-supervised-learning-problem-python/

And this:
https://machinelearningmastery.com/time-series-forecasting-supervised-learning/

REPLY

Sanjeev Kumar TM July 23, 2019 at 10:59 pm #

Hello,
Actually i trained the model with 4 predictors and later based on predictor importance one variable is not at all impact on response so i removed that parameter and trained the model but i am getting error obtained during 3 predictors is less as compared with 4 predictor model. will u please help me out why i am getting this error difference if i removed the parameter if it is not at all related to the response variable is reducing error or the error is same please help me out.

REPLY

Jason Brownlee July 24, 2019 at 7:55 am #

It is likely that the parameter that is "not useful" has nonlinear interactions with the other parameters and is in fact useful.

REPLY

Sanjeev Kumar TM July 23, 2019 at 11:00 pm #

How to get the coefficient of the predictor weights in ensemble boosted tree model.

REPLY

Jason Brownlee July 24, 2019 at 7:58 am #

You don't, they are not useful/interpretable.

REPLY

Priyanka Dave July 26, 2019 at 9:37 pm #

Hi Jason, I have total 47 input columns and 15 output columns (all are continuous values). Currently I am working on Random forest regression model.
My question is:

1) Can we define input -> output correlation or output -> output correlation ?
2) Can we tell model that particular these set of inputs are more powerful ?
3) Can we do sample wise classification ?
4) It is giving 98% accuracy on training data but still I am not getting expected result. Because model can not identify change in that particular input.

Thanks.

REPLY

Jason Brownlee July 27, 2019 at 6:09 am #

The algorithm will learn the relationships/correlations that are most relevant to making a prediction, no need to specify them.

You can make per-sample predictions, if you're using Python, here's an example:
https://machinelearningmastery.com/make-predictions-scikit-learn/

I recommend evaluating the model on a hold out test set, or better yet using cross validation:
https://machinelearningmastery.com/k-fold-cross-validation/

REPLY

Briana T August 7, 2019 at 2:12 am #

I just wanted to say that this explanation is so good and easy to follow! Thank you for providing this.

REPLY

Jason Brownlee August 7, 2019 at 8:02 am #

Thanks, I'm glad it helped!

REPLY

Luis Angel November 4, 2019 at 5:27 am #

Very thankful for your post.

I only have a simple question. The imbalanced sample could affect the performance of the algorithm?

REPLY

Jason Brownlee November 4, 2019 at 6:49 am #

Yes.

REPLY

Neha Manpreet November 16, 2019 at 1:25 am #

If bagging uses the entire feature space then in python we have max_features option in BaggingClassifier. Why we have this option of max_features ?

REPLY

Jason Brownlee November 16, 2019 at 7:26 am #

To limit the feature on each tree.

REPLY

Daniel January 4, 2020 at 7:02 pm #

Hi Jason,

I am confused on bootstrapping: how can we have a 'better mean' than the calculated one?

Why do I want to estimate the mean instead of calculating it?

THanks,
Daniel

REPLY

Jason Brownlee January 5, 2020 at 7:03 am #

A better estimate of the population mean from the data sample.

Recall that the population is all data, sample is a subset we actually have.

Perhaps see this tutorial:
https://machinelearningmastery.com/a-gentle-introduction-to-the-bootstrap-method/

REPLY

Daniel January 6, 2020 at 8:04 am #

Got it, thanks.

REPLY

Jason Brownlee January 6, 2020 at 1:30 pm #

You're welcome.

REPLY

Anil January 9, 2020 at 9:03 am #

Hello, Jason,
My question is;
– Does the random forest algorithm include bagging by default? So when I use the random forest algorithm, do I actually do bagging?
– If the random forest algorithm includes bagging by default and I apply bagging to my data set first and then use the random forest algorithm, can I get a higher success rate or a meaningful result?

REPLY

Jason Brownlee January 9, 2020 at 1:42 pm #

Yes and no. Think of it bagging by feature rather than by sample. You can also bag by sample by using a bootstrap sample for each tree.

REPLY

Paula García Llamas February 7, 2020 at 8:59 pm #

Hi Jason,

i am a bit confused with bagging in regression.

I run random forest with 1000 total observations, i set ntree to 1000 and i calculate the mean-squared error estimate and thus, the vaiance explained based on the out-of-bag.

If my ntree is 1000, that means that the number of bootstrap samples is 1000, each containing, by default, two thirds of the sampled poits and one third is used to get predictions out-of-bag, is this correct?

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Jason Brownlee February 8, 2020 at 7:10 am # REPLY

I am confused by your last statement.

The bootstrap samples are all different mixes of the original training dataset so you get full coverage.

samane February 25, 2020 at 8:29 pm # REPLY

Hi jason. thank u for complete explanation. I have a high dimensional data with few samples .() 47 samples and 4000 feature) is it good to use random forest for getting variable importance or going to Deep learning?

Jason Brownlee February 26, 2020 at 8:17 am # REPLY

I recommend testing a suite of different algorithms and discover what works best for your dataset.

Zamir March 31, 2020 at 1:53 pm # REPLY

Jason, thanks for your clear explanation.

I used 4 variables to predict one output variable. The importance analysis shows me that only one variable is useful. Then, I used random forest with this unique variable with good results.

Is it a correct approach and use of random forest? Is it also applicable for XGboosting?

Thanks,

Jason Brownlee April 1, 2020 at 5:44 am # REPLY

Perhaps try it and compare results?

Zamir April 21, 2020 at 7:11 am # REPLY

Thanks for your response,

Yes, both have similar results. Is it correct to use only one or two predictors for those machine learning models?

Jason Brownlee April 21, 2020 at 7:45 am # REPLY

Not sure about “correct”, use whatever gives the best results.

Larry Hertz May 4, 2020 at 9:11 pm # REPLY

Thank you Jason for this article !
Still I’m a little confuse with Bagging. If rows are extracted randomly with replacement, it is be possible that a feature’s value disappears from the final sample.
Hence, the associated decision tree might not be able to handle/predict data which contains this missing value.
How should a Random Forest model handle this case ? How to prevent it from such a situation ?

Jason Brownlee May 5, 2020 at 6:26 am # REPLY

No, because we create hundreds or thousands of trees and all data get a chance to contribute albeit probabilistically.

Stephen O June 4, 2020 at 11:14 am # REPLY

Hi Jason,

Thanks a bunch for the explanation.

1. Please I have about 152 wells. Each well has unique properties and has time series data with 1000 rows and 14 columns. I merged all the wells data to have 152,000 rows and 14 columns. I used the data for 2 wells for testing (2,000 rows and 14 columns). and the rest for training (2,000 rows and 14 columns). The random forest regression model performs well for training and poorly for testing and new unseen data. Please, what could be the issue?

2. Is it important to standardize before using random forest?

3. Can I specify the particular input variables/features to consider before splitting?

Please, help. Thanks a bunch

Jason Brownlee June 4, 2020 at 1:25 pm # REPLY

Perhaps some of the suggestions here will help:
https://machinelearningmastery.com/machine-learning-performance-improvement-cheat-sheet/

No, standardizing for RF won’t help, that’s my bet.

No need to specify features, RF will select the most appropriate features automatically.

Stephen O June 4, 2020 at 11:54 am # REPLY

Hello Jason,

Please, In what cases should we use BaggingRegressor (with a decision tree estimator) and in what cases should we use RandomForestRegreesor?

Thanks a bunch

Jason Brownlee June 4, 2020 at 1:26 pm # REPLY

There is no reliable mapping of algorithms to problems, instead we use controlled experiments to discover what works best.

Test both and use the one that is simpler and performs the best for your specific dataset.

Stephen O June 4, 2020 at 12:05 pm # REPLY

Hello Jason,

Please, pardon my name questions.

I am developing a model that considers all features before making a prediction. Should I use BaggingRegressor or RandomForestRegreesor?

Thank you.

Jason Brownlee June 4, 2020 at 1:26 pm # REPLY

Try it and see.

Val July 28, 2020 at 9:20 am # REPLY

Sir, your work is so wonderful and educative.Sir, Please I want to know how to plot mean square error against epoch using R.
Thanks so much for the work you are doing for us.

Jason Brownlee July 28, 2020 at 10:53 am # REPLY

Sorry, I don’t have an example of this in R.

Dinesh November 20, 2020 at 10:59 pm # REPLY

Sir,
Could You explain How the Sampling is done in random forest when bootstrap = True/False in sklearn?

Dinesh November 20, 2020 at 11:08 pm # REPLY

From above question,

I got to know that When Bootstrap is TRUE: Subsampling of Dataset (with sub rows and sub columns)

Bootstrap = False : Each tree considers all rows.
But what about sampling of columns for Bootstrap = False?

the sampling in the sense sampling of columns when Bootstrap =true/False

Jason Brownlee November 21, 2020 at 6:41 am # REPLY

The entire dataset (all rows) are used.

Jason Brownlee November 21, 2020 at 6:40 am # REPLY

When False, the whole dataset is taken I believe. When True, random samples with replacement are taken.

This is explained in the documentation here:
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html

vian February 10, 2021 at 9:44 pm # REPLY

hi, Mr.Jason
could you tell me the mathematical calculation behind feature important calculation using random forest. thx in advance.
regards.

Jason Brownlee February 11, 2021 at 5:55 am # REPLY

Not off hand, perhaps check a textbook or the paper.

Savitha March 31, 2021 at 8:37 pm # REPLY

Hello Jason,
I would like to know what is the difference between RandomForest vs BaggingClassifier(max_features < 1 , bootstrap_features = True) ie Subspace. In both these cases, we consider smaller set of features to derive the tree. Then what makes RandomForest special

Thanks

Jason Brownlee April 1, 2021 at 8:12 am # REPLY

Bagging will choose split points using all features, random forest uses a random subset of features.

Ajay A April 14, 2021 at 3:20 pm # REPLY

Hi jason, Does OBB prevents data Leakage?

Jason Brownlee April 15, 2021 at 5:22 am # REPLY

No. But it can be used in useful ways that avoid data leakage.

Genji April 14, 2021 at 5:43 pm # REPLY

Never miss a tutorial:

Jason Brownlee April 15, 2021 at 5:24 am #

REPLY

Thanks

Picked for you:

Leakey Mokaya September 24, 2021 at 12:01 pm #

REPLY

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For example, if a dataset had 25 input variables for a classification problem, then:

m = sqrt(25)

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Question: Is m = to 5 or 8 for the second m?

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Adrian Tam September 25, 2021 at 4:30 am #

REPLY

Why 8? You not 25 features and each split consider at most 5. That's what it means for the classification.

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reza December 16, 2021 at 5:15 am #

REPLY

Logistic Regression for Machine Learning

Hi Jason

I used the bagging algorithm but the results are the same for all data !! please guide me to replace the same alghorithm

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Adrian Tam December 17, 2021 at 7:19 am #

REPLY

Can you explain what you did?

Uche May 20, 2022 at 4:08 am #

REPLY

Hello Jason,

Your blog is a life saver . I have a dependent variable that is on a scale of 1-10. I also have 2 numerical independent variables and 10 independent variables on a linkert scale of 1-5. 4 of the independent variables are highly correlated when i ran correlation analysis. I also ran varaible importnace using RandomForest and found out that two of the highly correlated are the most important. My questions are these;

1) do i change the dependent variable that is on a scale of 1-10 to binary(dummy variable) of 1 and 0 or i should work with them without changing them?

2) to avoid multi-collinearity problem do i have to remove the hghly correlated variables according to least importance in the Random Forest and run the Random Forest again? or i should just conclude without taking out the least important variable that are highly correlated?

James Carmichael May 20, 2022 at 11:06 pm #

REPLY

Hi Uche…Thank you for the feedback and support! Please clarify some objectives for your model so that we may better assist you.

Leave a Reply

Name (required)

Email (will not be published) (required)

SUBMIT COMMENT

Welcome!
I'm *Jason Brownlee* PhD
and I **help developers** get results with **machine learning**.
Read more

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